Athrva Raut

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Education

Bharati Vidyapeeth (Deemed to be University) — Navi Mumbai, India
B.Tech in Artificial Intelligence and Machine Learning — *Aug 2024 – May 2028 (expected)*

- **CGPA:** 8.0/10
- **Coursework:** Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, Data Mining

Experience

Coders Technologies — Sep 2026 – Present
Full Stack Intern (sole developer of PackersMart) — *Navi Mumbai, India*

- Sole developer of **PackersMart**, an online marketplace connecting customers with packers-and-movers companies in the Mumbai region; responsible for backend, frontend, database and deployment
- Built with FastAPI, SQLAlchemy 2.0, Alembic, PostgreSQL and HTMX; deployed with Docker Compose and Caddy on a Linux server
- Wrote **260+ automated tests** (pytest) plus end-to-end browser tests (Playwright) at phone and desktop widths

Shalimar Nutrients Pvt. Ltd. — Apr 2026 – Jul 2026
Software Developer (part-time) — *Remote*

- Built a web app that replaced handwritten boiler and power logbooks at a 3-shift soya plant; operators logged 20+ readings per shift from their phones
- FastAPI + PostgreSQL (Supabase) backend with JWT auth and 3-role access control; React/TypeScript frontend
- Report engine (openpyxl) that rebuilt the plant's daily and monthly boiler, coal and power Excel reports
- Operator safeguards: one entry per shift slot, out-of-range warnings, and coal stock carried over day to day

Projects

Facial Expression Recognition | *TensorFlow, Keras, OpenCV, pytest* — Oct 2026

- Raised FER-2013 test accuracy from 54.05% to **69.96%**: found 7.9% of test images leaked into training, removed 1,798 bad images, matched training and webcam preprocessing (+4.3 points)
- Swapped the Haar detector for YuNet (faces found 72.9% to 96.7%); cut inference from 80 to 10 ms per face

GasSetu: Real-Time LPG Fraud Detection API | *FastAPI, scikit-learn* — Apr 2026 – May 2026

- Fail-fast check on every LPG booking: GSTIN format, live GSTN lookup, name matching and an Isolation Forest; SHA-256 hash-chained delivery log built and tested
- 94.2% accuracy (F1 0.85) on 10,000 synthetic records, tested against simple baselines to show what synthetic data cannot prove; deployed as a public API

NSE Algorithmic Trading Engine | *Python, AWS EC2, Angel One SmartAPI* — Mar 2026 – Present

- Swing-trading engine for 113 NSE stocks, 24/7 on AWS EC2 (paper trading): ATR sizing, idempotent orders
- Out-of-sample backtests on 2020–2025 data exposed overfitting (profit factor 1.01 in-sample vs 0.87 out-of-sample)

Pathfinder 3D | *Python, Flask, JavaScript, WebGL* — Mar 2026

- Races Dijkstra, A*, BFS, bidirectional search and D* Lite on a 3D grid with millisecond timing; routes between real cities with a TSP solver and caches repeated routes

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript
Web: FastAPI, SQLAlchemy, Alembic, Pydantic, PostgreSQL, REST APIs, JWT, HTMX, React, Tailwind CSS
Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, pandas, NumPy
DevOps & Tools: Docker, Linux, Caddy, AWS EC2, Supabase, Vercel, Git, pytest, Playwright, Claude Code
Certification: Machine Learning Specialization, DeepLearning.AI & Stanford (Coursera)